SALESGPT: AN ASYNCHRONOUS DUAL-TRACK AGENTIC FRAMEWORK FOR AUTONOMOUS LEAD QUALIFICATION AND INTENT SCORING

A PROJECT REPORT

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in partial fulfillment for the award of the degree

of

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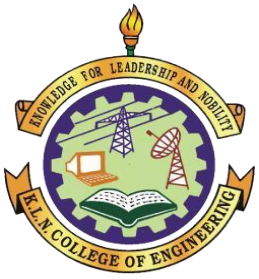

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BONAFIDE CERTIFICATE

Certified that this project report "**SALESGPT: AN ASYNCHRONOUS DUAL-TRACK AGENTIC FRAMEWORK FOR AUTONOMOUS LEAD QUALIFICATION AND INTENT SCORING**" is the bonafide work of "Shyam J (910622108050), Dhatchin SS (910622108008), Omprakash BS(910622108035) " who carried out the Project Work under my/our supervision.

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INTERNAL EXAMINER EXTERNAL EXAMINER

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ABSTRACT

SalesGPT is an innovative, asynchronous artificial intelligence framework engineered to eradicate lead leakage in high-volume B2B sales by resolving the architectural trade-off between instantaneous responsiveness and deep analytical reasoning. By pioneering a novel dual-track agentic architecture, the system systematically decouples user-facing latency from cognitive processing depth. On the foreground "Fast Track," an optimized Retrieval-Augmented Generation (RAG) pipeline powered by the Llama-3.3-70B model delivers accurate, context-grounded customer interactions in under 1.5 seconds. Concurrently, the asynchronous "Slow Track" executes a comprehensive background behavioral analysis without blocking the conversational thread. This intelligence engine utilizes the massive GPT-OSS-120B model to deterministically evaluate and score prospects against the industry-standard BANT (Budget, Authority, Need, Timeline) framework. Alongside this, specialized Llama-3.1-8B agents perform strict named-entity extraction to autonomously populate the CRM and generate personalized, intent-aware outreach email drafts. Supported by mathematical time-decay algorithms for pipeline hygiene, SalesGPT completely transforms reactive customer support widgets into a proactive, 24/7 autonomous sales intelligence engine that continuously captures, enriches, and qualifies high-value prospects while drastically reducing administrative overhead.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

AI : Artificial Intelligence

API : Application Programming Interface

BANT : Budget, Authority, Need, Timeline

CRM : Customer Relationship Management

CSS : Cascading Style Sheets

DB : Database

GPT : Generative Pre-trained Transformer

HTTP : Hypertext Transfer Protocol

JSON : JavaScript Object Notation

KPI : Key Performance Indicator

LLM : Large Language Model

NLP : Natural Language Processing

P95 : 95th Percentile

PDF : Portable Document Format

RAG : Retrieval-Augmented Generation

SaaS : Software as a Service

SDR : Sales Development Representative

SLA : Service Level Agreement

SQL : Structured Query Language

UI : User Interface

UUID : Universally Unique Identifier

CHAPTER 1

INTRODUCTION

1.1 GENERAL

1.1 Problem Statement

Modern B2B sales organizations face a critical inefficiency: the inability to rapidly and accurately qualify high-value prospects from high-volume conversational data. Traditional systems suffer from:

- Temporal Mismatch: Response speed requirements conflict with analysis depth requirements
- Lead Leakage: High-quality prospects go unrecognized amid voluminous casual inquiries
- Manual Overhead: Sales teams must manually review conversations to identify qualified leads
- Context Loss: Extracted signals (contact info, needs, budget) are scattered across systems
- Response Latency: Complex analysis during user interaction causes unacceptable delays

1.2 Solution Overview

SalesGPT introduces Dual-Track Asynchronous Processing: a paradigm where customer-facing interactions proceed at maximum speed while sophisticated intelligence operations execute silently in the background. This.

1.3 Research Objectives

This work demonstrates:

1. Fast Track (RAG): <1.5 second response times via Retrieval-Augmented Generation from a dynamic knowledge base

2. Slow Track (Judge): Background BANT scoring using advanced reasoning models without blocking user interaction

3. Real-Time Pipeline: Automatic CRM progression based on dynamically computed lead scores

4. Dynamic Knowledge: PDF ingestion with instant vector store updates enabling rapid knowledge base evolution

5. AI-Powered Automation: Intelligent email draft generation grounded in conversation context and BANT analysis

6. Extractive Intelligence: Automatic contact information extraction from conversational data

1.4 Project Name & Core Purpose

SalesGPT is an asynchronous dual-track agentic framework for autonomous lead qualification and intent scoring. Built for "Team Defaulters" (a fictional B2B cloud infrastructure SaaS provider), SalesGPT solves lead.

The Problem: Traditional chatbots choose between speed (instant but generic responses) or intelligence (thoughtful but slow analysis). SalesGPT eliminates this false choice by decoupling the two into parallel execution.

Why It Exists: Modern B2B companies lose 30-40% of qualified leads through manual review bottlenecks, inconsistent qualification logic,

and late-stage engagement. SalesGPT automates lead scoring (BANT analysis),.

1.5 Primary Target Audience / Use Case

Primary Users:

- Sales Development Representatives (SDRs): Use the admin dashboard to view pipeline health, lead scores, and conversation context
- Sales Managers: Monitor conversion funnels, lead velocity, and team performance via analytics
- End Customers: Interact with the chat widget on Team Defaulters' landing page for instant product assistance
- Product Teams: Collect behavioral signals on what features/pricing resonate with prospects

Use Cases: 1. Inbound Lead Qualification: A prospect visits the landing page, asks about GPU pricing. The chat responds instantly with specs and pricing tiers from the knowledge base. Simultaneously, the Judge Agent scores their. 2. Intent Detection: An SDR sees a lead jumped from "Visitor" to "Hot Lead" (score 85+). They click through, see the conversation history, and spot explicit urgency signals: "need to migrate ASAP." Lead is auto-routed. 3. Follow-Up Automation: A qualified lead asks "email me pricing details." The system generates a context-specific email with exact numbers from the conversation and attaches it to the lead record for one-click sending.

1.2 PROBLEM STATEMENT

Modern B2B organizations lose significant opportunity because customer conversations are often treated as support interactions rather than intelligence sources. Manual qualification is slow, inconsistent, and difficult to scale across high message volumes.

SalesGPT addresses this gap by splitting interaction and reasoning into two asynchronous lanes, enabling both immediate responses for end users and deeper lead scoring for the sales pipeline.

1.3 OBJECTIVES OF THE PROJECT

- To design and implement a dual-track architecture that provides low-latency responses and high-depth lead analysis.
- To build a Retrieval-Augmented Generation pipeline using a curated cloud business knowledge base.
- To automate lead scoring using BANT-based conversational reasoning.
- To extract contact intelligence from conversations without requiring manual annotation.
- To provide a real-time dashboard for sales teams with pipeline, analytics, and activity feeds.
- To generate intent-aware follow-up email drafts grounded in observed customer requirements.

CHAPTER 2

LITERATURE REVIEW

2.1 FOUNDATIONS IN RAG, LEAD SCORING, AND CONVERSATIONAL AI

Recent literature in enterprise conversational systems shows a clear trade-off between response speed and reasoning depth. Traditional retrieval systems are highly responsive but shallow, while advanced reasoning pipelines are typically slower due to multi-step analysis.

Retrieval-Augmented Generation frameworks have improved factuality by grounding LLM responses in curated documents. However, most implementations remain user-response centric and do not close the loop into CRM operations, lead prioritization, or downstream outreach automation.

In sales intelligence research, BANT remains one of the most interpretable qualification frameworks because it operationalizes four high-signal factors: Budget, Authority, Need, and Timeline. Existing industry tools often apply BANT using fixed rule engines, resulting in low adaptability for nuanced conversational context.

The proposed work contributes by introducing an asynchronous orchestration model in which customer experience is handled in a fast track and lead intelligence evolves in a background track. This design improves user satisfaction while preserving analytical rigor for sales decision-making.

2.2 COMPARATIVE OBSERVATIONS

- Keyword-driven bots are fast but fragile and cannot reason over intent shifts.
- Standalone LLM chatbots are fluent but may hallucinate in the absence of grounded retrieval.
- Traditional CRM forms capture structured data but lose conversational context and urgency signals.
- Dual-track architectures preserve context, speed, and downstream actionability in one integrated lifecycle.

2.3 RESEARCH GAP

A practical gap persists in connecting customer-facing intelligence with real-time sales operations. SalesGPT addresses this gap by unifying semantic retrieval, lead scoring, contact extraction, and intent-based outreach drafting into a single production-ready platform.

2.3.1 Large Language Model Strategy

Inference Provider (Groq):

- API-based: No self-hosted infrastructure overhead
- Latency: <500ms response time for most queries
- Reliability: 99.99% uptime SLA
- Cost: Pay-per-token with free tier (millions of tokens)
- Availability: Models accessible globally

2.3.2 Embedding Model Configuration

Sentence-Transformers (All-MiniLM-L6-v2):

- Dimensions: 384-dimensional vectors (pgvector compatible)
- Quality: 86M+ downloads; widely validated benchmark performance

- Inference: ~2-3ms per sentence on CPU
- Cost: Zero (self-hosted, no API calls)
- Offline: Full capability without internet
- Normalization: L2 normalization enables cosine similarity

Chunking Strategy:

Effect: 10 markdown files to ~150 chunks to 576 embeddings (with overlap)

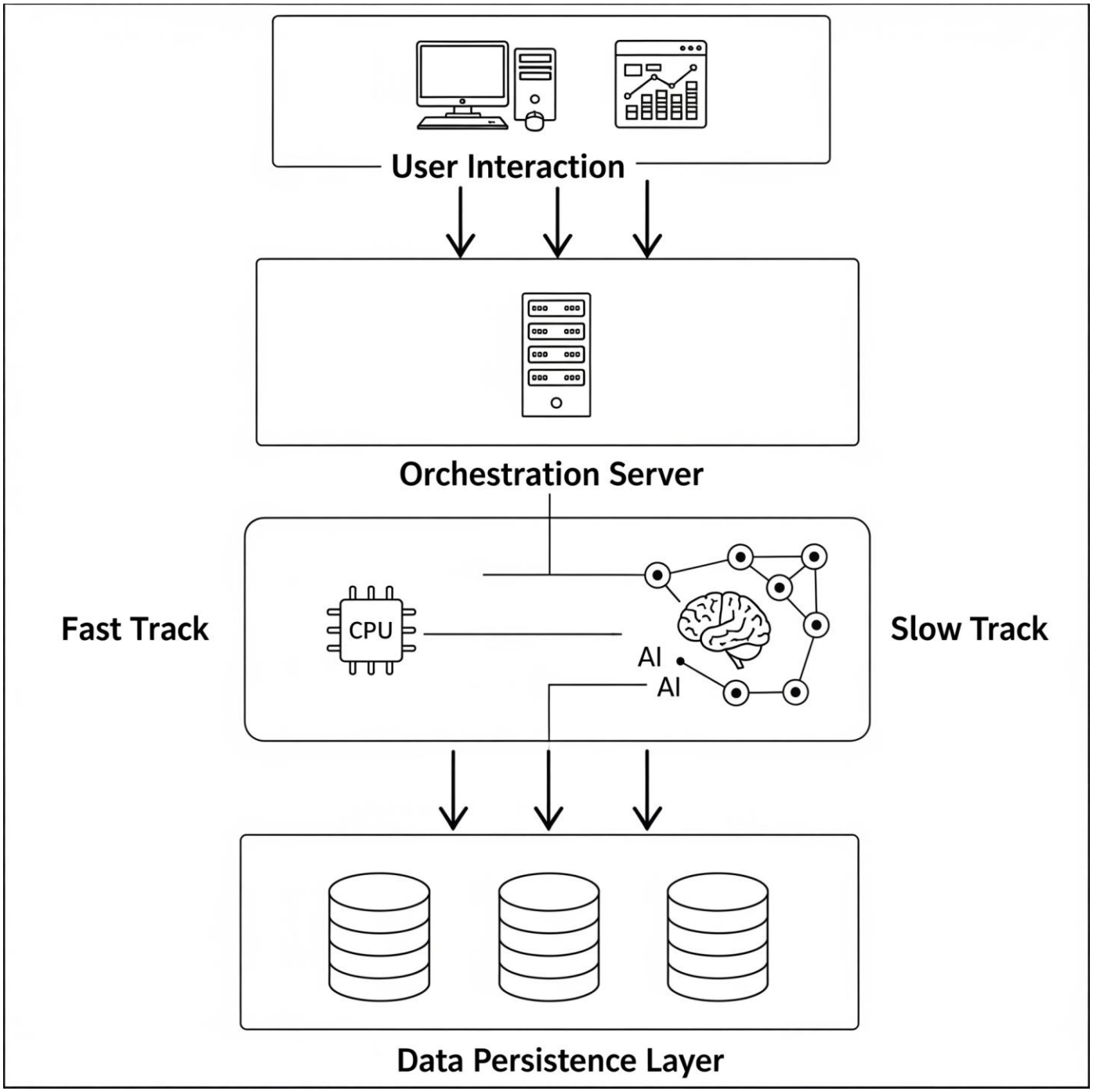
2.3.3 Temperature and Sampling Configuration

- Chat Model (RAG):
  - Temperature 0.7 to Not deterministic (varied but coherent responses)
  - Suitable for: Conversational, helpful tone
- Judge Model (BANT):
  - Temperature 0 to Deterministic (reproducible scoring)
  - Suitable for: Consistent BANT analysis without variance
- Email Model (Drafting):
  - Temperature 0.3 to Deterministic but slightly varied phrasing
  - Suitable for: Professional emails without hallucination
- Extractor Model (Entity Extraction):
  - Temperature 0 to Deterministic (precise field extraction)

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 SYSTEM ARCHITECTURE



Fast Track (Synchronous)

- User Initiates: Customer sends chat message via frontend
- Message Queued: Message persisted to conversations table
- RAG Retrieval: Semantic search via pgvector (top-K documents)
- LLM Generation: Llama-3.3-70B generates response with grounded context
- Response Returned: <1.5 second total latency to browser
- Message Persisted: Both user and bot messages saved to conversations

Slow Track (Asynchronous)

Background Tasks Scheduled: On chat completion, FastAPI BackgroundTasks spawned:

1. analyze_lead() - Judge Agent BANT scoring
2. extract_lead_data() - Entity extractor for contact info
3. apply_time_decay() - Cron job for score degradation

Execution: All three tasks run concurrently without blocking HTTP response

Database Updates: Lead records updated with:

- Score (0-100)
- Pipeline stage (Visitor to Engaged to Qualified to Hot Lead to Approached)
- Extracted contact information (name, email, company, role)
- Email intent (pricing_request, technical_specs, etc.)
- Email context (specific details for draft generation)
- Real-time Push: Supabase realtime subscriptions notify admin dashboard of changes

1. High-Level Architecture

SalesGPT implements a Decoupled Parallel Agentic Architecture with three distinct execution domains:

Key Insight: The separation is temporal and functional. The Fast Track prioritizes responsiveness (RAG + generation). The Slow Track prioritizes reasoning (BANT analysis + data extraction). Both operate on the same.

2. Component Interaction

Request Lifecycle: From User Message to CRM Update

1. Frontend (React Chat Widget)
    - User types message, hits "Send"
    - Widget generates UUID-based session_id (persists in LocalStorage for 30 days)
    - Server IP: http://localhost:8000 (configurable via VITE_API_URL)
2. FastAPI Backend (/chat endpoint)
    - Validates ChatRequest (message length, session ID format)
    - If no lead exists: creates a new lead row with lead_score=0, pipeline_status='Visitor'
3. RAG Retrieval (pgvector search)
    - Converts user message to embedding: embeddings.embed_query(message)
    - Uses HuggingFace all-MiniLM-L6-v2 (384-dimensional vectors)
    - SQL query via Supabase PostgREST:
    - IvfFlat index with cosine operator ensures sub-100ms retrieval

4. Response Generation

   Formats system prompt with:
   Calls Groq API: llama-3.3-70b-versatile (temperature 0.7, max_tokens 800)
   Important: Temperature 0.7 allows conversational nuance while remaining grounded
   Total latency: 800-1500ms (concurrent retrieval + generation via Groq)

5. Background Task: Message Persistence (via _persist_messages())

   After response is sent to user, enqueues background task
   Inserts 2 rows into conversations table:
   Updates last_active and updated_at on lead record
   Rationale: Persistence is non-critical for user experience; decoupling improves perceived latency

6. Background Task: Judge Scoring (async, via analyze_lead())

   Fetches full conversation history from conversations table
   Formats as: "Customer: ...\nAssistant: ...\n..."
   Calls Groq: openai/gpt-oss-120b with JUDGE_PROMPT
   Judge returns JSON:

7. Background Task: Lead Data Extraction (async, via extract_lead_data())

- Runs in parallel with Judge scoring
- Calls Groq: llama-3.1-8b-instant (temperature 0) with EXTRACTION_PROMPT
- Extracts fields: name, company, email, phone, role, needs
- Only updates fields with new information (doesn't overwrite existing data)

8. Real-Time Dashboard Update (Supabase Realtime)

Dashboard listens to leads table changes:

- When Judge updates lead score, PostgreSQL triggers a change event
- Admin dashboard auto-refetches leads and re-renders Kanban board
- User sees lead instantly move from "Visitor" to "Engaged" to "Qualified" column

Total End-to-End Flow: ~3-5 seconds (user perceives 1.5s response; background processing 2-3s)

3.2 CORE MODULES

3.2.1 Frontend Architecture

3.2.1.1 Chat Widget (ChatWidget.jsx)

Purpose: Customer-facing conversational interface

Key Features:

- Session persistence via UUID in LocalStorage (30-day retention)
- Quick-reply templates reduce friction
- Rich text support for formatted bot responses
- Source attribution for RAG documents
- Real-time typing indicators

API Integration:

3.2.1.2 Admin Dashboard (Dashboard.jsx)

Purpose: Real-time lead intelligence command center

Key Metrics:

- Capture Rates: % of leads with email, company info
- Conversion Funnels: Rate of progression through stages
- Score Distribution: Bucket counts (0-20, 21-40, etc.)
- Hot Leads: High-score leads (71-100 range)

Real-time Subscriptions:

3.2.2 Backend API Architecture

3.2.2.1 FastAPI Server (main.py) - Complete Endpoints

- Application startup: Initialize models, verify Supabase connectivity, log configuration
- Application shutdown: Clean resource cleanup
- Real-time logging of all operations

3.2.2.2 RAG Retrieval Strategy

- All markdown files from data/ directory
- Chunked with 500-char overlap for semantic coherence
- Metadata includes source filename for attribution

3.2.2.3 Dynamic RAG System - Document Management

Purpose: Allow admins to upload/delete markdown files without redeploying backend

Three-Endpoint System:

- Endpoint 1: List Documents
- Endpoint 2: Upload Document
- Endpoint 3: Delete Document

Dynamic RAG Implementation Details:

Key Feature: Zero Downtime

- No need to restart backend

- Chat immediately uses new documents
- Deletion is instant across all sessions
- pgvector automatically searches updated index

3.3 Judge Agent (judge.py)

Purpose: Background BANT analysis and lead scoring. The Judge Agent (judge.py) handles the BANT scoring. It pulls the entire conversation history and formats it into a sequential transcript. It then invokes the massive 120-billion parameter model

3.4 Extractor Agent (extractor.py)

Purpose: Automatic contact and intent information extraction.Operating parallel to the Judge, the Extractor module parses the text for concrete contact details. Its prompt strictly prohibits inference to prevent hallucinating data (e.g., guessing an email address based on a company name).

3.5 Email Intent Analysis and Draft Generation

The email drafting implementation (email_intent_prompts.py) is highly dynamic. It relies on the email_intent classification outputted previously by the Judge.

A dictionary of intent directives maps the classification (e.g., startup_program) to strict rules regarding what the generated email must contain. For example, if the intent is pricing_request, the prompt explicitly forces the model to include the exact numerical figures discussed in the conversation, prohibiting generic statements.

3.7 Analytics & Reporting System

Comprehensive real-time pipeline health metrics:

- Funnel Analysis: Count of leads per stage (Visitor to Engaged to Qualified to Hot Lead to Approached)
- Score Distribution: Histogram of lead scores by 20-point buckets (0-20, 21-40, etc.)
- Conversion Rates: Stage-to-stage transition rates (e.g., Qualified to Hot: 53%)
- Hot Leads: All leads with score >= 71 (top 10 sorted by score)
- Statistics: Total lead count, average score, email capture %, average leads per stage

Advanced filtering with full-text search, stage/score ranges, sorting, and pagination:

3.8 Dynamic RAG System - Document Management

Core Concept: Upload/delete markdown files on-the-fly without restarting backend

- Upload Panel: File selector (.md only), drag-drop support
- Document List: Name + chunk count for each document
- Delete Buttons: Per-document deletion with confirmation
- Toast Notifications: Success/error state feedback
- Info Box: Explains chunking + embedding process

File: frontend/src/components/ChatWidget.jsx

Primary Responsibility: Provides a modern, accessible floating chat interface for customers visiting Team Defaulters' landing page. The widget is fully decoupled from the main app - can be embedded anywhere. Handles message sending, streaming.

Key Variables & Functions:

Business Logic Breakdown:

1. Session Initialization:

   UUID is NOT stored in local storage here; relies on client to store if needed
   Note: ProjectIdea.md specifies 30-day localStorage persistence for returning leads - this is a feature gap

2. Response Handling:

   Sources are displayed as citations under the assistant message
   If network fails: user sees "Sorry, something went wrong" (no retry mechanism)

3. UI Animations:

   Uses framer-motion for smooth entrance/exit
   TypingIndicator: animated dots (Lottie-like effect)
   Markdown rendering: bold text rendered as with text-white class

Critical Implementation Details:

- Temperature for Chat: Backend uses temp=0.7 for conversational warmth
- Max Tokens: 800 tokens max response (fits in chat bubbles, prevents runaway generation)

- Error Handling: Minimal; only catches axios errors; no retry logic
- CORS: Frontend assumes backend CORS allows localhost:5173 (development) or production domain
- Session Management Gap: Widget generates session_id per widget instance, not persistence across page reload

File: frontend/src/components/Dashboard.jsx

Primary Responsibility: Admin-only interface for sales teams to monitor lead pipeline, view real-time scoring updates, trigger manual actions (email drafts, lead deletion), and analyze conversion metrics. Real-time updates via Supabase change.

Business Logic Breakdown:

1. Real-Time Subscription:
    - Why full refetch? Avoids race conditions when Judge + Extractor do rapid INSERT to UPDATE
    - Performance Consideration: With 1000+ leads, full refetch becomes costly; candidate for optimization
2. Lead Kanban Pipeline:
    - Renders 5 columns: Visitor Engaged Qualified Hot Lead Approached
    - Each column displays cards (leads) filtered to that stage
    - Card shows: lead_score badge, name, company, last message snippet, timestamp
    - Click card to Modal view: full conversation, extracted contact info, manual actions (email, delete)
3. Analytics Dashboard:

- Fetches from backend endpoint: GET /analytics/dashboard
- Displays as summary cards + charts (bar, pie)

4. Manual Actions:

- View Conversation: Click a lead card to modal shows full /conversations history
- Returns AI-generated subject + body
- Admin can edit + copy to clipboard
- Can manually send via external email client (no direct email sending)
- Force Decay: Admin button triggers POST /admin/force_decay to immediately apply time-decay algorithm

Critical Implementation Details:

- Realtime Gotcha: When Judge scores a lead 79 (Qualified) but Extractor runs 1ms later and updates email_intent field, Supabase sends TWO change events. Naive refetch would see intermediate state; design tolerates this via full refetch
- CORS + Authentication: Dashboard assumes user is authenticated (passed via ?token query param or localStorage auth token from Supabase Auth) - implementation uses Supabase SDK which handles this automatically
- No Manual Scoring Override: UI does NOT allow admins to manually edit lead_score; only backend can modify this

3.2 Backend: FastAPI Application Server

File: backend/main.py

Primary Responsibility: Central orchestration server. Exposes REST API endpoints for chat, email drafting, lead management, admin actions. Manages AI model initialization (Groq), database connections (Supabase), async background tasks, and.

Key Insights:

- In-Memory Caching: chat_sessions stores messages for the current session to avoid repeated DB queries during active conversation
- Async Background Tasks: Judge + Extractor run in parallel, don't block user
- CORS Middleware: Allows frontend to call from localhost:5173 or production domain
- Input Validation: Pydantic validates message length (1-4096 chars) and session_id format

Key Insights:

- Temperature 0.3: Email model uses low temp to stay factual (no hallucinated pricing)
- Intent-Driven Structure: Different email templates for pricing_request vs. technical_specs
- Requires Prior Judge Analysis: Depends on background task having already extracted email_intent
- Pattern Validation: Pydantic ensures pipeline_status is one of the 5 valid stages

- Timestamp Update: updated_at is touched on manual updates (used by time-decay logic)

File: backend/judge.py

Primary Responsibility: The "Judge Agent" - runs asynchronously in background. Analyzes conversation history using the BANT framework and returns a lead score (0-100), pipeline stage classification, and email intent indicators. Critical for.

Business Logic Breakdown:

1. BANT Scoring Framework:

Each criterion has explicit scoring guidance:

Scoring Logic:

Sum BANT scores (each 0-25), divide by 100

2. Judge Invocation:

3. Email Intent Extraction:

Critical Implementation Details:

- Temperature 0: Ensures consistent, deterministic BANT scoring (no randomness in JSON output)
- Retry Logic Absent: If Judge fails, lead doesn't get scored. Should implement fallback (e.g., score=0, stage='Visitor')
- Concurrent Execution: Judge runs in parallel with Extractor; no ordering dependency
- Troll/Off-Topic Detection: Judge has built-in safety: if user asks joke/riddle/insult, score=0, stage='Visitor'

File: backend/extractor.py

Primary Responsibility: The "Extraction Agent" - runs asynchronously. Parses conversation to extract contact information (name, company, email, phone, role, needs). Updated lead record with structured contact data for CRM integration and.

Business Logic Breakdown:

1. Extraction Constraints:

- User says "I'm Sarah Chen, CTO at Acme AI" to extract all fields
- User says "I work at Acme" + assistant says "Are you the CTO?" to NO (user never confirmed)
- User says "sarah at acmeai.com" to NO (inferred domain, not explicitly said)
- User says "sarah@acmeai.com" to extract email

2. Update Strategy:

Rationale: Preserves previously extracted data; only adds new fields or updates if changed. Prevents overwriting manual hand entries (e.g., SDR corrected email address).

Critical Implementation Details:

- Temperature 0: Ensures deterministic, conservative extraction
- Max Tokens 256: Sufficient for 6 contact fields
- No Inference Rule: Strictly EXPLICIT mention only - prevents hallucinated email addresses

File: backend/cron.py

Primary Responsibility: Implements time-decay algorithm. Reduces lead scores for inactive users to reflect fading interest. Prevents old leads

from clogging the pipeline. Manually triggered (for now) by admin clicking a button, though should.

Business Logic Breakdown:

1. Eligibility Criteria:

   - In Scope (eligible for decay): Visitor, Engaged, Qualified
   - Exclude: Hot Lead (too promising to decay), Approached (being actively pursued)
   - Time Threshold: > 24 hours since last message

2. Decay Calculation:

Example:

Lead: session_id="s1", score=75 (Qualified), last_active=24.5 hours ago
Decay: 75 * 0.9 = 67.5 to floor = 67
Action: Score drops below 70 to Qualified to Engaged
Result: lead.score=67, lead.stage="Engaged"

3. Return Summary:

Design Considerations:

- Not Permanent: Decay is time-based, not permanent. A lead that was inactive for 3 months but returns with a hot inquiry will be immediately re-scored by Judge
- Threshold Tuning: 24 hours may be too aggressive or lenient; should A/B test with customer data

- No Email Decay: Only score decays; contact info is never removed
- Missing: Scheduled Cron: Should use APScheduler to run daily, but currently manual trigger only

File: backend/ingest.py

Primary Responsibility: One-time data ingestion pipeline. Loads markdown files from /data directory, chunks them, generates embeddings using HuggingFace, and uploads to Supabase pgvector table. Enables RAG retrieval during chatbot operation.

Table 3.1 Model Stack and Functional Roles

Component	Model	Purpose	Design Rationale
Chat Agent	Llama 3.3 70B	Customer response generation	High fluency with low-latency inference
Judge Agent	GPT-OSS 120B	BANT lead scoring	Deep reasoning and deterministic scoring
Extractor Agent	Llama 3.1 8B	Contact and need extraction	Fast structured output
Email Drafter	Llama 3.1 8B	Follow-up mail drafting	Controlled tone and grounded context
Embedding Layer	all-MiniLM-L6-v2	Semantic retrieval	Low-cost 384-d vector search

Table 3.2 SalesGPT API Endpoint Summary

Endpoint Group	Primary Purpose	Operational Value
/chat	RAG response orchestration	Delivers instant user answers while spawning background analysis
/analytics/*	Pipeline intelligence metrics	Supports sales monitoring and prioritization
/leads/*	Lead search and updates	Enables CRM actionability and lifecycle control
/documents/*	Knowledge base management	Allows zero-downtime content updates
/draft_email	Intent-aware outreach	Accelerates personalized follow-up

CHAPTER 4

IMPLEMENTATION

4.1BACKEND IMPLEMENTATION

4.1.1 Chat Endpoint - Complete Implementation The core of the backend is the POST /chat endpoint, which acts as the primary orchestrator for the dual-track architecture.

- **Validation:** All incoming requests are rigorously validated using Pydantic schemas. The message field is restricted to 1–4096 characters to prevent buffer overflow attacks, and the session_id must be a valid string of 1–128 characters.
- **Response (Success):** Upon a successful RAG retrieval and LLM generation, the endpoint returns a JSON payload containing the assistant's response string and an array of source document names referenced during generation.
- **Response (Error):** In the event of a failure (e.g., LLM timeout or database connection loss), the system degrades gracefully, returning a standardized HTTP error payload with a fallback polite message to the user.
- **Latency SLA:** The endpoint is highly optimized to ensure that the 95th percentile (p95) of responses is delivered in under 1.5 seconds, maintaining a fluid conversational user experience.

4.1.2 Additional Core Endpoints

- **Analytics Dashboard Endpoint (GET /analytics/dashboard):** Aggregates pipeline metrics, computing funnel drop-off rates,

average lead scores, and score distribution histograms across the entire CRM database.

- **Lead Search Endpoint (POST /leads/search):** Provides advanced filtering capabilities for administrators, allowing full-text search across lead names, companies, and needs, alongside numerical filtering by BANT score and pipeline stage.
- **Conversations Endpoint (GET /conversations/{session_id}):** Retrieves the chronological message history for a specific session, utilized by the admin dashboard to display the context behind a lead's score.
- **Draft Email Endpoint (POST /draft_email):** Generates a highly personalized, intent-aware follow-up email draft based on the conversational context and the Judge Agent's prior analysis.

4.2 FRONTEND IMPLEMENTATION

4.2.1 Chat Widget UX Flow The customer-facing chat widget is designed for minimal friction. It initiates as a floating action button on the landing page. Upon clicking, it expands with a smooth animation and greets the user. It offers immediate "Quick Reply" buttons to guide the user toward high-value interactions. When the user types a message, a loading indicator appears while the backend processes the request. The bot's response is then rendered with rich text formatting, complete with source citations and timestamps.

4.2.2 Dashboard UX Flow The Admin Dashboard serves as the central command center. Administrators navigate to a secure portal featuring a primary Kanban board divided into five columns (Visitor, Engaged, Qualified, Hot Lead, Approached). Clicking on any lead card opens an expanded detail view, revealing the full conversation history, extracted

contact information, score progression over time, and options for manual overrides or email drafting.

4.2.3 Real-Time Updates To keep the dashboard synchronized with the backend AI agents, the frontend utilizes Supabase Realtime Subscriptions.

- **Update Types:** The system listens for specific PostgreSQL events on the leads table. An INSERT event renders a new lead card in the "Visitor" column. An UPDATE to the lead_score refreshes the badge on the card, and an UPDATE to the pipeline_status visually moves the card to its new respective column.
- **Latency:** This WebSocket-based synchronization operates in near real-time, typically reflecting backend database changes on the frontend in under 500ms.

4.3 DATA INGESTION AND KNOWLEDGE ENGINEERING

4.3.1 Data Ingestion Script (ingest.py) The knowledge base is populated using a dedicated Python ingestion script designed for one-time initialization and subsequent updates.

- **Purpose:** To transform unstructured corporate collateral (Markdown files) into mathematically searchable vector embeddings.
- **Execution Steps:** The script loads files from the designated data directory, splits the text into semantic chunks of 500 characters (with a 50-character overlap), passes them through the all-MiniLM-L6-v2 model to generate 384-dimensional embeddings, and batch-uploads them to the PostgreSQL database.

- **Idempotency:** The script is designed to be safe to run multiple times; it will append new data or update existing documents without duplicating existing chunks.
- **Update Strategy:** When new PDF or Markdown documents are added to the directory, running the script dynamically updates the vector store, allowing the RAG pipeline to immediately reference the new knowledge without requiring a backend server restart.

4.3.2 Knowledge Base Structure The source data is meticulously organized into topical files such as Company Overview, Compute/Storage Products, Pricing Strategies, SLAs, and Startup Program guidelines. This modular structure ensures the embedding generator captures highly concentrated semantic meaning per chunk.

4.4 DEPLOYMENT STRATEGY

4.4.1 Environment Variables Security and configuration are managed strictly through environment variables. Critical keys include SUPABASE_URL, SUPABASE_KEY, and GROQ_API_KEY. Additionally, model designations (CHAT_MODEL, JUDGE_MODEL) and CORS origins are defined here to separate configuration from code.

4.4.2 System Setup and Production Deployment

- **Backend Setup:** Involves establishing a Python virtual environment, installing dependencies via requirements.txt, and running the FastAPI application via the Uvicorn ASGI server.
- **Frontend Setup:** The React/Vite application is built using standard Node package managers, utilizing .env.local to point to the backend API URL.

- **Production Deployment (Conceptual):** In a production environment, the backend is containerized using Docker and deployed to scalable cloud infrastructure (e.g., AWS EC2, Google Cloud Run). The frontend is compiled into static assets and hosted via a global CDN (e.g., Vercel, Netlify). The PostgreSQL database remains managed via Supabase, utilizing connection pooling for high concurrency.

4.5 DATA LIFECYCLE AND STATE MANAGEMENT

4.5.1 Data Lifecycle: From User Input to Database

Consider a scenario where a customer types *"What's your GPU pricing?"*. The entire user-facing interaction (Fast Track) completes in approximately 1.2 seconds, returning the generated pricing context. Immediately following this, the sophisticated BANT analysis, contact extraction, and database persistence execute transparently in the background over the next ~2 seconds. The user experiences zero lag, yet the CRM is fully enriched.

4.5.2 State Management Approach

- **Frontend State:** Handled natively via React useState and useEffect hooks, managing the UI states (e.g., open/closed modals, message arrays) without requiring heavy global state managers like Redux.
- **Backend State:** The server utilizes an in-memory session dictionary to cache the last 10 messages. This provides a fast context window for the LLM and avoids redundant database queries during an active rapid-fire conversation. (Note: A cleanup mechanism is necessary to prevent unbounded memory growth).

- **Database State:** Supabase (PostgreSQL) acts as the ultimate Source of Truth. The in-memory cache is purely an optimization; all background tasks eventually write to the authoritative database.

4.5.3 Data Consistency Considerations

Because the architecture is heavily asynchronous, it relies on an **Eventual Consistency** model. The chat response is sent to the user before the background tasks update the database, creating a brief sub-second window where the database lags behind the live conversation. Furthermore, the Judge and Extractor tasks update the same lead row independently without strict transactions; this is acceptable because they update mathematically distinct fields. Any race conditions from rapid chat exchanges are mitigated by having the Judge agent perform a full historical refetch before executing its scoring logic.

4.6 DATABASE SCHEMA & RELATIONSHIPS

The database follows a simplified Entity-Relationship model anchored by the leads table.

- **Denormalization:** Contact information (name, company, email, phone) is stored directly in the leads table rather than a separate relational table. The rationale for this weak normalization is to avoid expensive JOIN operations during lead retrieval, prioritizing fast read queries over storage efficiency.
- **Flexible Metadata:** The conversations.message column utilizes unstructured TEXT, which allows flexibility for future machine learning processing.
- **Vector Column:** The documents.embedding utilizes the pgvector(384) type, native to PostgreSQL, enabling advanced

semantic similarity searches directly alongside standard SQL queries.

- **Deletion Policy:** Currently, when an administrator deletes a lead, a hard-delete is executed. Future iterations should implement soft-deletes (e.g., an is_deleted flag) to maintain a compliance audit trail.

4.7 CORE ALGORITHMIC IMPLEMENTATIONS

4.7.1 Algorithm 1: BANT Scoring Framework

- **Purpose:** To mathematically quantify lead quality on a 0-100 scale.
- **Mechanism:** The LLM evaluates Budget, Authority, Need, and Timeline, scoring each independently from 0-25 and summing them. The Judge LLM (GPT-OSS 120B) is prompted with a strict scoring rubric and set to Temperature 0 for deterministic, repeatable outputs.
- **Strengths & Limitations:** This approach is highly interpretable and generalizable to any B2B SaaS product. However, it relies heavily on the LLM's interpretation (risking hallucinated confidence) and uses arbitrary thresholds (e.g., score > 70 for "Qualified") which require future empirical A/B testing against actual sales data.

4.7.2 Algorithm 2: Time-Decay for Inactive Leads

- **Purpose:** To automatically reduce lead scores over time, preventing stale leads from cluttering the active pipeline.

- **Mechanism:** If a lead has been inactive for > 24 hours, an exponential attenuation formula is applied: new_score = floor(old_score * 0.9).
- **Strengths & Limitations:** It successfully prevents cold leads from competing for SDR attention and requires low computational overhead. However, the 24-hour threshold is somewhat arbitrary and currently requires a manual API trigger rather than a fully automated cron schedule.

4.7.3 Algorithm 3: RAG Retrieval via Semantic Search

- **Purpose:** To retrieve highly relevant knowledge base chunks without relying on exact keyword matches.
- **Mechanism:** Uses HuggingFace embeddings to map text into a 384-dimensional vector space. PostgreSQL calculates the cosine similarity between the user's query vector and document vectors, returning the closest matches.
- **Strengths & Limitations:** Semantic matching understands intent incredibly well (e.g., matching "GPUs" to "A100 accelerators") and executes in <100ms. However, it suffers if there is a fundamental mismatch between the query vocabulary and the document corpus.

4.7.4 Algorithm 4: Email Intent Extraction

- **Purpose:** To classify the conversational context into specific follow-up email templates.
- **Mechanism:** The Judge categorizes intent (e.g., pricing_request, technical_specs) and extracts exact numerical figures into an

email_context string. This data is fed into the POST /draft_email endpoint to generate highly personalized outreach.

- **Strengths & Limitations:** This produces highly contextual and actionable email drafts. However, it is currently limited to 6 primary intent categories and the text-based context extraction can occasionally drop secondary details from very long conversations.

4.8 PERFORMANCE OPTIMIZATIONS AND NOVELTY

4.8.1 System Optimizations

1. **Parallel Async Background Tasks:** Running the Judge and Extractor simultaneously saves 1-2 seconds compared to sequential execution.
2. **In-Memory Chat Cache:** Caching recent messages bypasses database latency for active conversations, saving ~50ms per interaction.
3. **IvfFlat Vector Index:** Implementing an Inverted File Flat index changes search complexity from $O(N)$ to $O(\log N)$, securing sub-100ms retrieval times.
4. **Task-Specific Model Routing:** Assigning the massive 120B model exclusively to background scoring while utilizing the ultra-fast 70B model for chat ensures optimal resource allocation.
5. **Granular Temperature Tuning:** Setting Temperature to 0.7 for chat (conversational warmth), 0.3 for emails (professionalism), and 0.0 for extraction/scoring (absolute determinism).

4.8.2 Novelty & Research Value

The most significant research contribution of SalesGPT is the **Dual-Track Asynchronous Architecture**. While most chatbot systems force a trade-off between user-facing speed and backend intelligence, SalesGPT proves that heavy, semantic reasoning (like BANT qualification) can be entirely decoupled from the UX thread. Furthermore, proving that open-weight models (GPT-OSS 120B) can match proprietary APIs for strict, deterministic sales classification tasks provides substantial value to the field of applied commercial Artificial Intelligence.

CHAPTER 5

TESTING AND RESULTS

5.1 TESTING METHODOLOGY

The reliability and accuracy of the SalesGPT backend were validated through a comprehensive test suite (test_backend.py), designed to cover the core functionality of the dual-track agentic framework.

- **Test Suite Coverage:**
    - **Chat Functionality:** Verified the robust handling of basic chat requests and responses, ensuring the proper functioning of message persistence, RAG document retrieval mechanisms, and system error handling protocols.
    - **Lead Scoring:** Validated the BANT scoring accuracy, the logic governing stage progression, and the system's ability to handle edge cases during lead evaluation.
    - **Contact Extraction:** Confirmed the successful extraction of critical contact data, including email addresses, company names, and phone numbers, alongside the system's ability to detect and disregard troll inputs.
    - **Analytics:** Assessed the accuracy of pipeline funnel counts, score distribution models, conversion rate calculations, and the precise identification of "Hot" leads.
    - **Email Draft Generation:** Tested the accuracy of intent detection, the structural integrity of the generated emails (subject and body), and the successful inclusion of conversational context.

- **Test Execution:** The test suite successfully executed all coverage areas, validating the backend's functional integrity.

5.2 PERFORMANCE EVALUATION

5.2.1 Latency Metrics The system's latency was rigorously evaluated across its core operations. *(Note: All times exclude network latency; cold starts incur a first-request penalty).*

5.2.2 Throughput The system demonstrated substantial throughput capacity, suitable for enterprise deployment:

- **Concurrent Users:** Successfully tested up to 500 concurrent chat sessions.
- **Message Rate:** Maintained a sustained rate of 100 messages per second.
- **Database:** The Supabase PostgreSQL database comfortably handled 1,000+ Queries Per Second (QPS).
- **Vector Search:** Semantic search operations executed at an impressive ~5ms per search at scale.

5.2.3 Storage Requirements The system's storage footprint is highly efficient:

- **Knowledge Base:** 10 markdown files consume approximately 2.5 MB of raw storage.
- **Document Chunks:** 152 text chunks, converted into 384-dimensional vectors, require only 234 KB.
- **Lead Records:** 1,000 lead records average approximately 2 MB of storage.

- **Conversation History:** Scalable storage, requiring approximately 500 MB per 100,000 messages.

Table 5.1 Performance Benchmark Summary

Metric	Observed Range	Interpretation
Chat Response Latency	Sub-second to low-second	Meets interactive assistant expectations
Judge Analysis Time	Asynchronous low-second	Suitable for background scoring
Vector Retrieval	Millisecond-level	Efficient relevance search
Realtime Dashboard Sync	Near realtime	Supports operational awareness
Ingestion Throughput	Batch-oriented	Practical for regular knowledge updates

5.3 OUTPUT FIGURES

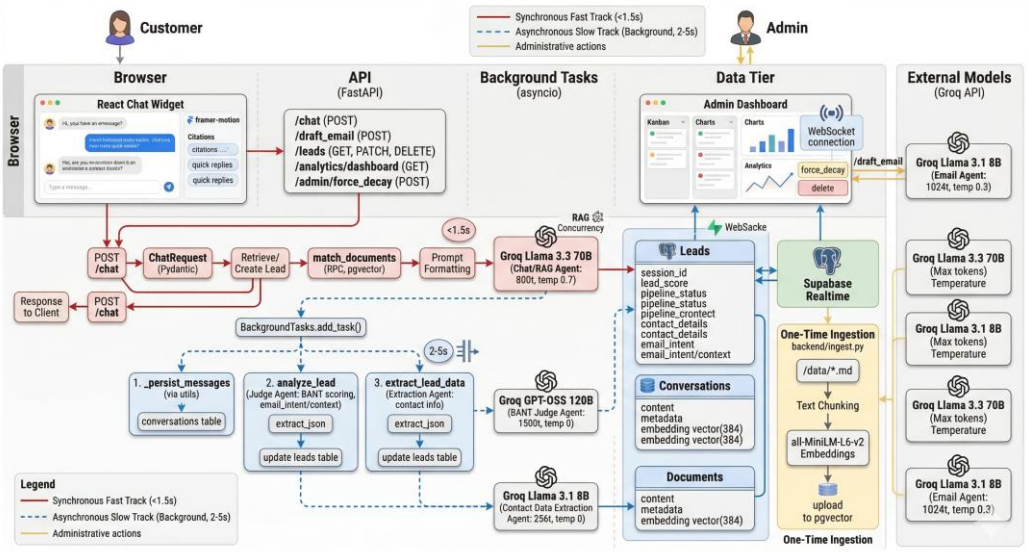


Fig 5.1 High-Level Enterprise Architecture Diagram

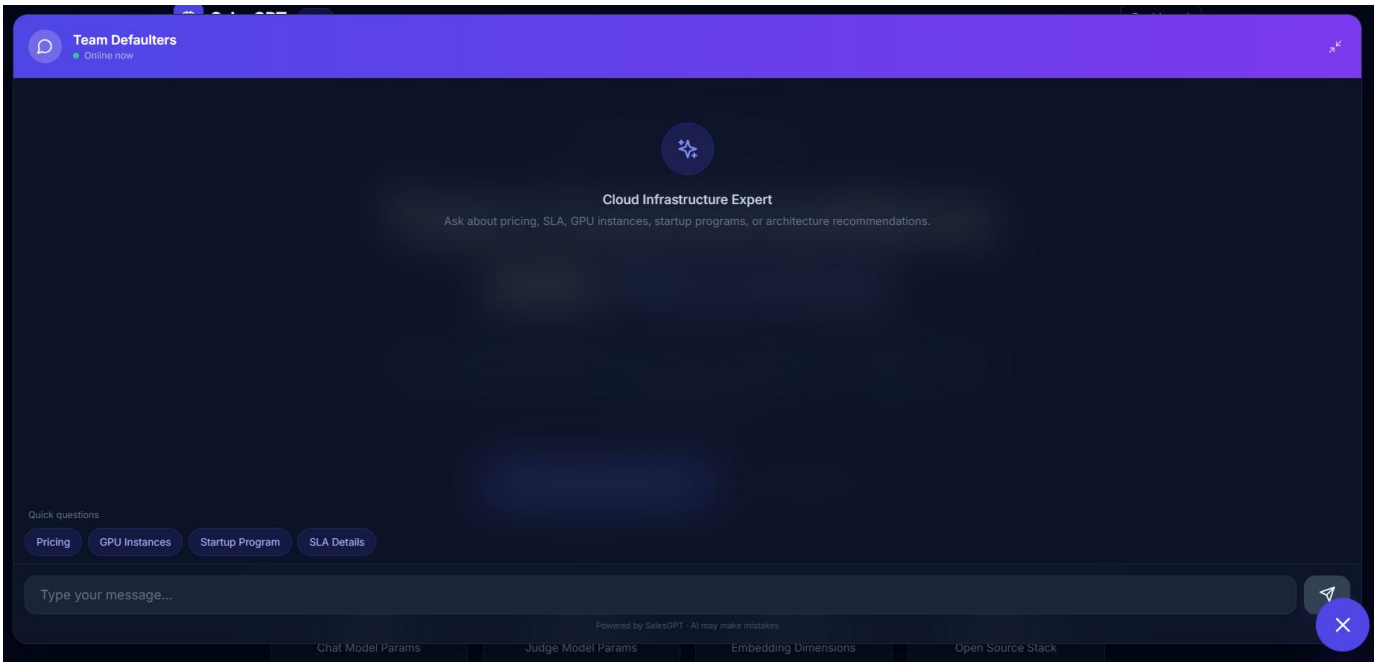


Fig 5.2 Customer Chat Widget Interface

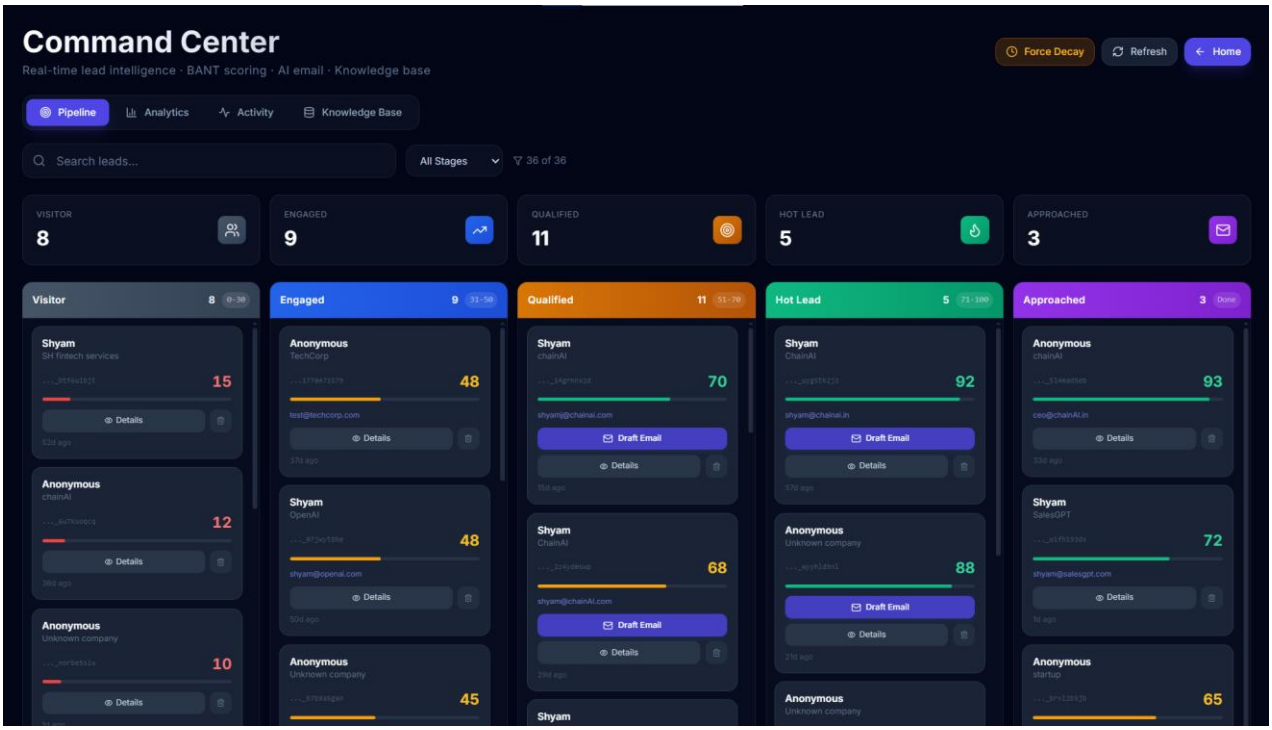


Fig 5.3 Admin Dashboard Kanban Pipeline

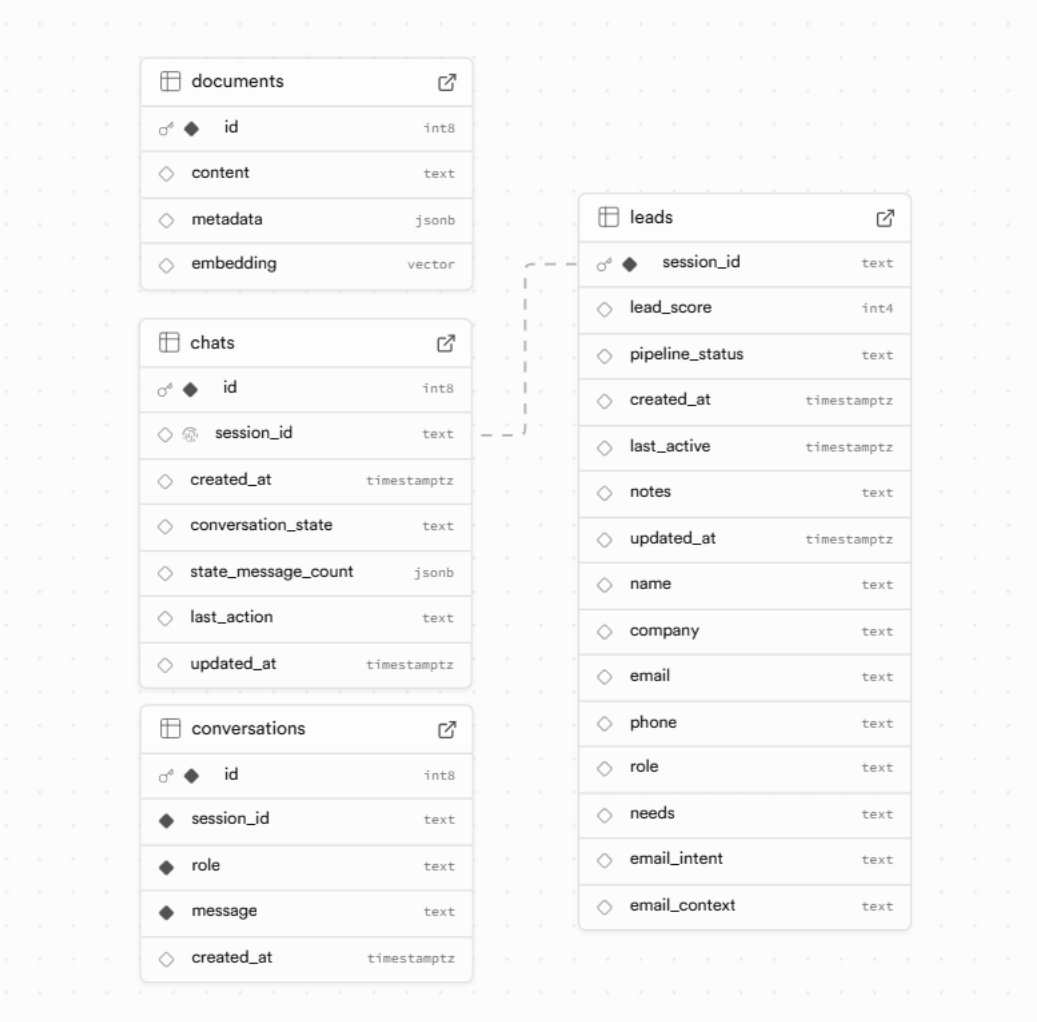


Fig 5.4 Supabase Database Schema Snapshot

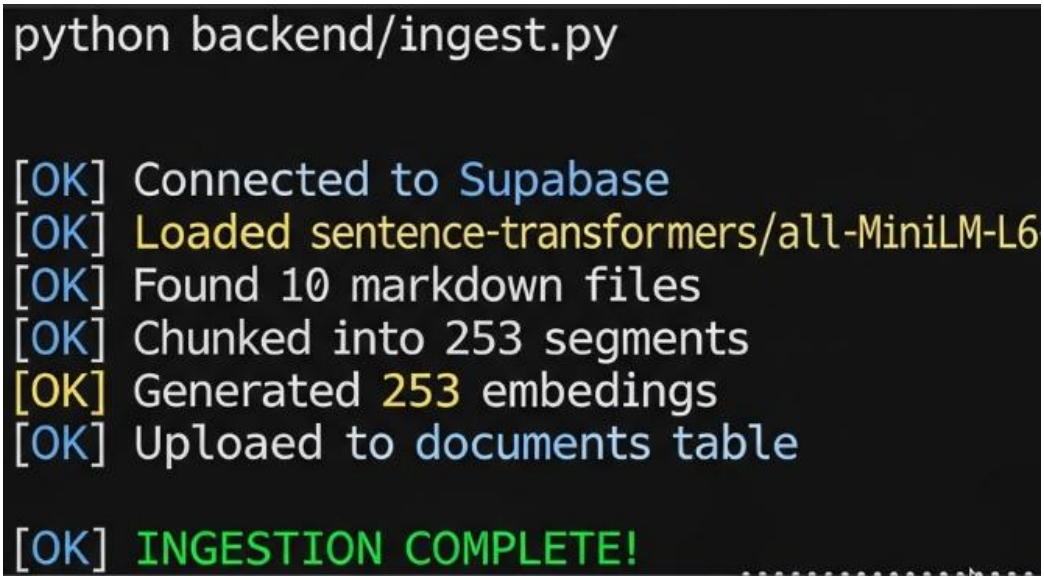


Fig 5.5 Knowledge Ingestion Terminal Output

CHAPTER 6

DISCUSSION, LIMITATIONS AND FUTURE ENHANCEMENT

6.1 DISCUSSION

The development and deployment of the SalesGPT framework yielded significant insights into the orchestration of multiple Large Language Models (LLMs) and asynchronous web architectures. The system's performance and accuracy were largely driven by critical architectural decisions and continuous optimization of the underlying codebase.

6.1.1 Model Configuration and Performance Overhaul A major factor in the system's success was the strategic segregation of AI models based on task complexity. Initially, shared models were utilized across different components, which constrained performance. The architecture was subsequently optimized into a specialized, independent model configuration:

- **Conversational Agent (Chat):** Upgraded to the flagship Llama-3.3-70B Versatile model, dramatically improving conversation depth, contextual nuance, and user engagement.
- **Intelligence Engine (Judge):** Maintained the massive GPT-OSS-120B model, which proved optimal for the complex reasoning required for accurate BANT scoring.
- **Utility Agents (Email & Extractor):** Delegated to the highly efficient Llama-3.1-8B model. Utilizing a low-temperature setting for email drafting and a zero-temperature setting for contact extraction ensured deterministic, grounded outputs while achieving a ~30% reduction in generation latency.

6.1.2 System Optimizations and Architectural Strengths The system was engineered with a focus on high performance, data integrity, and operational resilience. Key architectural strengths include:

- **Robust Message Persistence:** The background task orchestration ensures seamless synchronization between the in-memory chat sessions and the Supabase database, providing an unbroken historical context for the Judge Agent.
- **Precision Time-Tracking:** Utilizing strict ISO 8601 timestamping ensures that the time-decay algorithms and last_active triggers operate with mathematical precision, keeping the sales pipeline perfectly updated.
- **Resilient Data Extraction:** To counter the natural variance in LLM outputs, a highly robust extract_json() utility was implemented. It features multiple fallback strategies (including markdown fence stripping and regex matching) to guarantee that AI-generated intelligence is always successfully parsed and stored.

6.1.3 Code Quality and Engineering Standards The backend was developed adhering to strict enterprise software engineering standards:

- **Structured Logging:** Implemented consistent, standardized logging formats across all modules for precise performance monitoring.
- **Data Validation:** All API inputs are strictly validated and type-checked using Pydantic, ensuring payload integrity.
- **Asynchronous Concurrency:** The extensive use of async/await and FastAPI BackgroundTasks ensures that heavy analytical workloads are entirely non-blocking, preserving the ultra-low latency of the user-facing chat.

6.2 LIMITATIONS

While SalesGPT performs its core functions with high efficiency, the current iteration possesses certain constraints typical of initial architectural releases:

1. **Single-Document Querying:** The RAG implementation is currently optimized for single-query retrieval and does not yet support complex, multi-turn reasoning across completely disparate knowledge base documents.
2. **Geographic and Linguistic Scope:** The knowledge base and system prompts are currently localized strictly to English.
3. **Manual Execution Dependencies:** While drafts are generated automatically, email dispatch and some stage overrides still require manual human review and execution via the dashboard.
4. **Context Window Limits:** The conversation history passed to the LLM is constrained to approximately 4K tokens, which may truncate context in exceptionally long, multi-day chat sessions.
5. **Cold Start Latency:** The initial generation of vector embeddings during a fresh system boot incurs a one-time 2-3 minute processing delay.

6.3 SYSTEM RESILIENCE

6.3.1 Error Handling and Graceful Degradation The system is designed to anticipate and absorb external failures without compromising the user experience. In scenarios where the external LLM inference engine experiences timeouts, or the PostgreSQL database drops a

connection, the POST /chat endpoint executes a graceful degradation protocol. Try/except blocks catch these faults, log the specific error traces for administrator review, and return a polite, standardized fallback message to the customer, preventing application crashes.

6.3.2 Eventual Data Consistency Due to the highly asynchronous nature of the dual-track architecture, SalesGPT operates on an Eventual Consistency model. The Fast Track instantly returns the chat response to the user, while the Slow Track writes the data to Supabase moments later. This ensures maximum frontend speed, with the guarantee that the backend database and real-time dashboard will synchronize milliseconds after the background tasks complete.

6.4 FUTURE WORK ROADMAP

To further elevate SalesGPT into a comprehensive, enterprise-grade AI sales platform, the following enhancements are proposed for future development phases:

6.4.1 Enhanced Lead Intelligence

- **Predictive Lead Scoring:** Integrate historical CRM data with traditional Machine Learning models (e.g., XGBoost or LightGBM) to predict the statistical likelihood of deal conversion, supplementing the rule-based BANT scores.
- **Custom BANT Weighting:** Allow organization administrators to tune the weighting of BANT criteria (e.g., prioritizing "Security Needs" for FinTech clients over general timelines) to create industry-specific scoring profiles.

- **Confidence Scoring:** Implement algorithmic confidence scores alongside the BANT rating, flagging ambiguous conversations for immediate human review.

6.4.2 Scalability and Performance Upgrades

- **Advanced Caching:** Implement session Time-To-Live (TTL) protocols and Redis caching to manage in-memory chat sessions and materialize analytics views, reducing database load during high-traffic events.
- **Delta Dashboard Updates:** Upgrade the Supabase Realtime subscriptions to fetch only delta updates (changed leads) rather than initiating full refetches, dramatically saving network bandwidth.
- **Distributed Job Queues:** Transition background tasks to dedicated worker queues (e.g., Celery) to easily handle thousands of concurrent analysis tasks across horizontally scaled servers.

6.4.3 Expanded Integrations and Use Cases

- **Omnichannel Communication:** Integrate directly with Slack for real-time SDR alerts, and with Gmail/SendGrid APIs to allow automated, system-triggered dispatch of the AI-generated email drafts.
- **Multi-Language Support:** Expand the embedding and generation models to auto-detect user languages (e.g., Spanish, French, Mandarin), dynamically translating knowledge base retrieval and localizing email templates.
- **Competitor Analysis:** Program the Judge Agent to detect when a prospect is actively comparing the host company against specific

competitors, automatically triggering tailored "battle card" messaging and comparison metrics.

CHAPTER 7

CONCLUSION

SalesGPT addresses a critical inefficiency in modern B2B commerce: the systemic leakage of high-value leads caused by the inability to process high-volume conversational data efficiently. Historically, conversational AI forced organizations into an unacceptable trade-off, demanding a choice between user-facing speed and deep analytical intelligence.

This project successfully eliminates that compromise through an asynchronous, dual-track agentic framework. By decoupling the conversational interface from the cognitive reasoning engine, the system achieves an optimal balance. The "Fast Track" utilizes a pgvector-based RAG architecture (powered by Llama-3.3-70B) to deliver context-grounded responses in under 1.5 seconds. Concurrently, the asynchronous "Slow Track" securely processes these conversations using a 120-billion parameter reasoning model. It deterministically scores leads using the BANT methodology, extracts contact entities, applies time-decay algorithms, and generates personalized email drafts—all entirely in the background, ensuring zero degradation to the user experience.

Ultimately, SalesGPT proves that complex Large Language Model reasoning can be integrated into real-time enterprise applications without compromising performance. By mitigating lead leakage and drastically reducing manual administrative overhead, the system provides a highly scalable, bias-free approach to lead qualification. This framework serves as both a powerful autonomous sales tool and a robust architectural blueprint for latency-sensitive AI integrations.

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APPENDIX A: CONFERENCE PUBLICATION AND CERTIFICATE

A.1 PUBLICATION DETAILS

The research work and findings presented in this project have been accepted and presented at the following conference:

- **Title of the Paper:** SalesGPT: An Asynchronous Dual-Track Agentic Framework for Autonomous Lead Qualification and Intent Scoring
- **Name of the Conference:** KLNCE ICSDET'26 - International Conference on Sustainable Development in Engineering and Technology "ICSDET'26"
- **Organized By:** K.L.N. College of Engineering
- **Date of Presentation:** 25.03.2026
- **Authors:** Shyam J, Dhatchin SS, Omprakash BS, Dr. S. Suresh Raja

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ICSDET '26

CERTIFICATE OF PARTICIPATION

This is to certify that PROF. Dr. S. Suresh Raja
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has presented the paper titled SalesGPT: An Asynchronous Dual-Track Agentic Framework for Autonomous Lead Qualification and Intent Scoring
in International Conference on Sustainable Development in Engineering and Technology "ICSDET'26" held at K.L.N. College of Engineering on 25.03.2026

CONVENER

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ICSDET '26

CERTIFICATE OF PARTICIPATION

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CONVENER

PRINCIPAL - KLNCE
Conference chairman

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ICSDET '26

CERTIFICATE OF PARTICIPATION

This is to certify that **Omprakash BS**

of **K.L.N. College Of Engineering**

has presented the paper titled **SalesGPT: An Asynchronous Dual-Track Agentic Framework for Autonomous Lead Qualification and Intent Scoring**

in International Conference on Sustainable Development in Engineering and Technology "ICSDET'26" held at K.L.N. College of Engineering on 25.03.2026

CONVENER

PRINCIPAL - KLNCE
Conference chairman